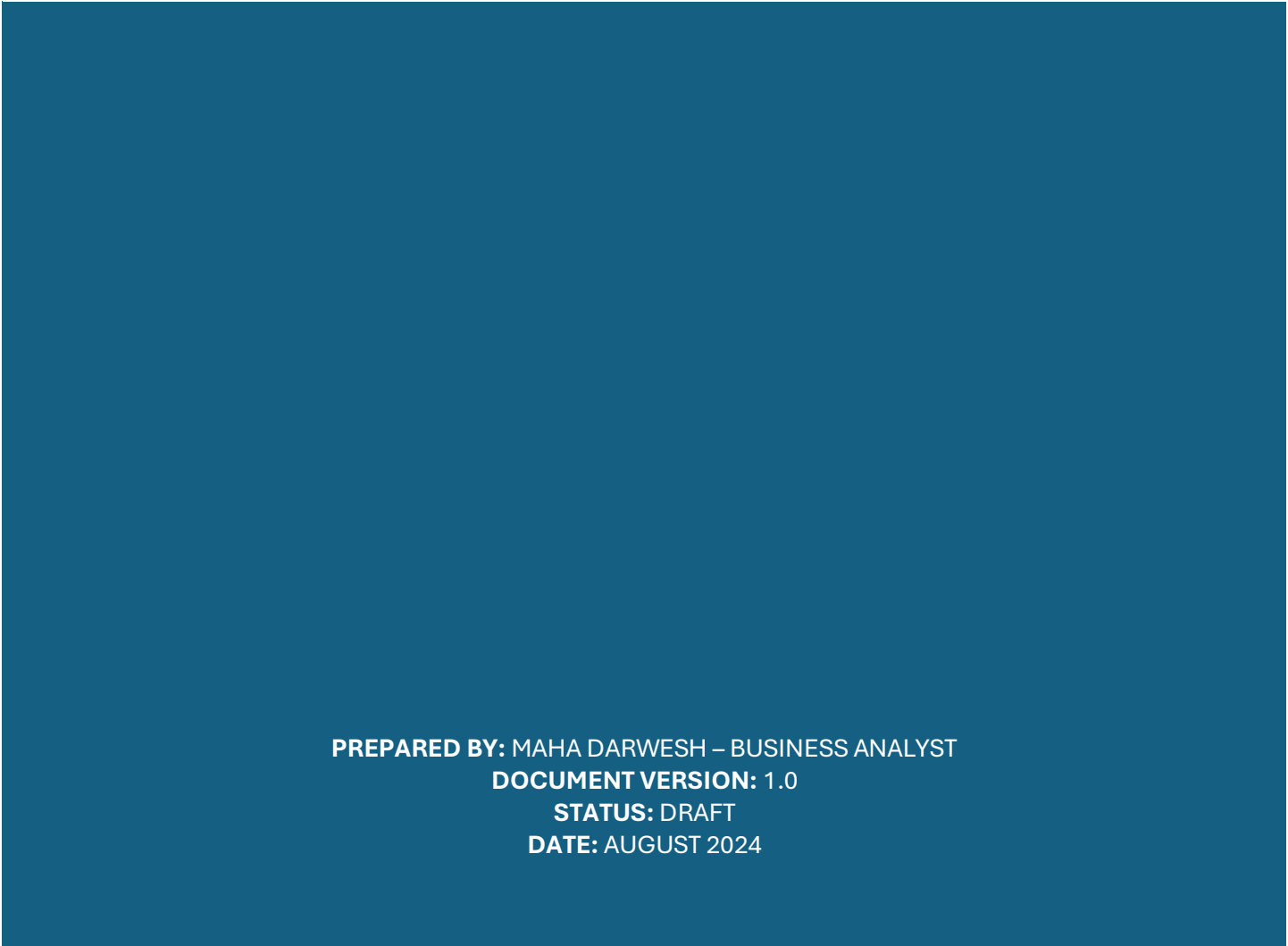




USE CASE SPECIFICATION

STUDENT REGISTRATION SYSTEM



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Document Purpose & Use

This master specification consolidates the approved working set of UC-001 through UC-029 for the Student Registration System Enterprise Edition.

Use cases represent end-to-end business interactions and controls rather than a one-to-one rewrite of functional requirements. Traceability therefore allows one use case to reference multiple FRs and one FR to participate in multiple use cases.

The specification is intended to support SRS review, process modeling, acceptance-test design, and the Requirements Traceability Matrix (RTM).

Specification Conventions

- Fail closed means that a mandatory security/authorization/authoritative-data failure does not grant access or fabricate a business decision.

- Atomic/transactional means that commit-critical state changes succeed together or roll back to a consistent state.
- Idempotent means that retrying the same business request does not create duplicate side effects.
- Effective-dated rules retain the policy/version needed to explain historical decisions.

Use Case Index

ID	Use Case	Primary FR Focus
UC-001	Student Login & Authentication	FR-001, FR-005, FR-006, FR-008, FR-009, FR-010, FR-031
UC-002	Forgot Password / Password Recovery	FR-002, FR-005, FR-008, FR-027, FR-031
UC-003	Change Password	FR-003, FR-008, FR-010, FR-027, FR-031
UC-004	Secure User Logout	FR-004, FR-006, FR-007, FR-009, FR-031
UC-005	Session Timeout & Automatic Logout	FR-004, FR-006, FR-031
UC-006	Remember Me / Trusted Login	FR-001, FR-004, FR-005, FR-006, FR-007, FR-008, FR-031
UC-007	Multi-Factor / Step-Up Authentication	FR-001, FR-008, FR-010, FR-031
UC-008	Concurrent Session Management	FR-004, FR-006, FR-009, FR-031
UC-009	Role-Based Access Control (RBAC)	FR-008, FR-010, FR-031
UC-010	Student Course Registration	FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019, FR-027, FR-031, FR-033, FR-034
UC-011	Course Search & Section Availability	FR-012, FR-018, FR-030, FR-034
UC-012	Registration Eligibility Validation	FR-013, FR-014, FR-015, FR-016, FR-017, FR-029, FR-033, FR-034

ID	Use Case	Primary FR Focus
UC-013	Seat Allocation & Concurrent Registration	FR-011, FR-018, FR-019, FR-031
UC-014	Add / Drop / Withdraw Course	FR-020, FR-022, FR-023, FR-024, FR-027, FR-034
UC-015	Waitlist Management	FR-018, FR-021, FR-027, FR-034
UC-016	Schedule & Registration Status Synchronization	FR-011, FR-020, FR-021, FR-022, FR-031, FR-035
UC-017	Tuition & Fee Calculation	FR-020, FR-023, FR-026, FR-029, FR-034
UC-018	Invoice & Student Account Posting	FR-023, FR-024, FR-026, FR-031, FR-035
UC-019	Online Payment Processing	FR-024, FR-025, FR-027, FR-031, FR-035
UC-020	Scholarships, Waivers & Discounts	FR-023, FR-024, FR-026, FR-028, FR-031
UC-021	Notifications & Communications	FR-027, FR-031, FR-035
UC-022	Exception & Approval Workflow	FR-008, FR-010, FR-028, FR-029, FR-031
UC-023	Academic Policy & Rule Administration	FR-010, FR-013, FR-014, FR-015, FR-016, FR-017, FR-028, FR-029, FR-031
UC-024	Course Section & Offering Administration	FR-010, FR-012, FR-018, FR-021, FR-027, FR-029, FR-030, FR-031, FR-034, FR-035
UC-025	Audit Trail & Compliance Monitoring	FR-001–FR-035; FR-031 primary
UC-026	Reporting & Management Dashboards	FR-010, FR-018, FR-021, FR-023, FR-026, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035
UC-027	Student Profile & Academic Status Management	FR-010, FR-011, FR-016, FR-020, FR-023, FR-026, FR-028,

ID	Use Case	Primary FR Focus
		FR-029, FR-030, FR-031, FR-032, FR-033, FR-035
UC-028	Academic Term & Calendar Management	FR-010, FR-011, FR-012, FR-017, FR-020, FR-021, FR-022, FR-023, FR-027, FR-029, FR-030, FR-031, FR-032, FR-034, FR-035
UC-029	Enterprise Integration & Data Synchronization	FR-019, FR-022, FR-024, FR-025, FR-027, FR-031, FR-033, FR-035

UC-001 – Student Login & Authentication

Use Case ID	UC-001
Primary Actor(s)	Student / Authorized User
Supporting Systems / Actors	Authentication Service; User Database; Audit Service
Trigger	User selects Login.
Preconditions	User account exists and is active; authentication service is available.
Related Functional Requirements	FR-001, FR-005, FR-006, FR-008, FR-009, FR-010, FR-031
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Display the secure login interface.
2	Accept university credentials over an encrypted channel.
3	Validate the credentials and account status.
4	Evaluate failed-attempt and account-lockout controls.
5	Invoke MFA when policy requires it.
6	Evaluate concurrent-session policy.
7	Create a secure server-side session after successful authentication.
8	Resolve the user role and permissions through RBAC.
9	Record the authentication event in the audit trail.
10	Redirect the user to the authorized landing area.

Alternative / Exception Flows

- Invalid credentials: deny access without revealing sensitive account details and record the failed attempt.
- Account is locked/inactive: deny authentication.
- MFA fails or expires: do not create an authenticated session.
- Authentication/authorization dependency is unavailable: fail closed.

- Concurrent-session policy is exceeded: apply the configured session control.

Postconditions

- A valid authenticated session exists only after all required controls succeed.
- No protected resource is available after a failed authentication attempt.

Security & Enterprise Controls

- HTTPS/TLS; secure session cookies; server-side authorization; MFA/step-up where required; rate limiting/lockout; auditable authentication events.

UC-002 – Forgot Password / Password Recovery

Use Case ID	UC-002
Primary Actor(s)	Student / Authorized User
Supporting Systems / Actors	Authentication Service; Identity Verification Service; Notification Service; Audit Service
Trigger	User selects Forgot Password.
Preconditions	A registered account may exist; recovery services are available.
Related Functional Requirements	FR-002, FR-005, FR-008, FR-027, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Display the password-recovery interface.
2	Accept the permitted account identifier.
3	Return a neutral response that does not disclose whether an account exists.
4	Perform the configured identity verification.
5	Generate a cryptographically secure, single-use, time-limited reset token.
6	Deliver recovery instructions through the approved channel.
7	Validate the token when the user opens the recovery link.
8	Accept and validate the new password against password policy.
9	Change the password and invalidate the reset token.
10	Invalidate existing sessions/tokens when required by policy.
11	Send a security notification and record the event.

Alternative / Exception Flows

- Unknown identifier: return the same neutral response used for a known account.
- Expired/used/invalid token: reject and allow a new recovery request.
- Identity verification fails: stop recovery.

- New password violates policy: reject with policy guidance.
- Excessive recovery attempts: apply rate limiting and security monitoring.
- Required recovery service is unavailable: make no credential change.

Postconditions

- On success, the new password is active and the reset token cannot be reused.
- On failure, existing credentials remain unchanged.

Security & Enterprise Controls

- Single-use expiring tokens; account-enumeration resistance; rate limiting; encrypted transport; audit logging.

UC-003 – Change Password

Use Case ID	UC-003
Primary Actor(s)	Authenticated User
Supporting Systems / Actors	Authentication Service; User Database; Notification Service; Audit Service
Trigger	User selects Change Password.
Preconditions	User is authenticated and the account is active.
Related Functional Requirements	FR-003, FR-008, FR-010, FR-027, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Open account-security settings.
2	Request current password, new password, and confirmation.
3	Verify the current credential.
4	Apply step-up MFA when policy requires it.
5	Validate the new password against password policy and confirmation.
6	Update the password securely.
7	Invalidate affected sessions/tokens according to policy.
8	Record the change and send a security notification.
9	Confirm successful completion to the user.

Alternative / Exception Flows

- Current password is incorrect: reject the change.
- New password violates policy or confirmation differs: reject without changing credentials.
- Step-up MFA fails: cancel the operation.
- Session is expired: require re-authentication.
- Authentication service is unavailable: fail closed.

Postconditions

- The new credential becomes effective only after successful validation.
- A failed attempt leaves the current password unchanged.

Security & Enterprise Controls

- Step-up authentication for sensitive changes; no plaintext password storage; secure transport; audit and notification.

UC-004 – Secure User Logout

Use Case ID	UC-004
Primary Actor(s)	Authenticated User
Supporting Systems / Actors	Session Management Service; Authentication Service; Audit Service
Trigger	User selects Logout.
Preconditions	A session exists or the client presents session state.
Related Functional Requirements	FR-004, FR-006, FR-007, FR-009, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Receive the logout request.
2	Identify the current authenticated session.
3	Invalidate the session server-side.
4	Invalidate the corresponding session token/cookie.
5	Remove sensitive client authentication state.
6	Record the logout event.
7	Redirect the user to the login page.
8	Require fresh authentication for subsequent protected requests.

Alternative / Exception Flows

- Session is already expired: clear local state and redirect to Login.
- Invalid token is presented: deny access and clear client session state.
- User chooses Logout All Sessions: invalidate all permitted active sessions and persistent tokens.

Postconditions

- The terminated session cannot be reused.

- Protected resources require a new successful authentication.

Security & Enterprise Controls

- Server-side invalidation; Secure/HttpOnly cookies; anti-cache controls for sensitive pages; audit logging.

UC-005 – Session Timeout & Automatic Logout

Use Case ID	UC-005
Primary Actor(s)	Authenticated User
Supporting Systems / Actors	Session Management Service; Audit Service
Trigger	Configured inactivity threshold is approached or reached.
Preconditions	User has an authenticated session.
Related Functional Requirements	FR-004, FR-006, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Track inactivity using trusted server-side session state.
2	When the configured warning point is reached, display a clear timeout warning.
3	Provide Continue Session and Logout options.
4	If Continue Session is selected in time, validate the session before extending it.
5	If Logout is selected, terminate the session immediately.
6	If no response is received during the two-minute warning period, terminate the session automatically.
7	Invalidate the session and tokens server-side.
8	Redirect to Login and record the timeout event.

Alternative / Exception Flows

- Session expires while the warning is visible: do not extend it; require login.
- Activity arrives after expiry: reject it and do not reactivate the expired session.
- Session-state validation is unavailable: fail closed.

Postconditions

- Expired sessions cannot access protected resources.

- A new authentication is required after timeout.

Security & Enterprise Controls

- Server-side timeout enforcement; two-minute warning; non-reusable expired tokens; auditable timeout events.

UC-006 – Remember Me / Trusted Login

Use Case ID	UC-006
Primary Actor(s)	Authorized User
Supporting Systems / Actors	Authentication Service; Session Management Service; Audit Service
Trigger	User selects Remember Me during authentication.
Preconditions	User can complete normal authentication; feature is permitted by policy.
Related Functional Requirements	FR-001, FR-004, FR-005, FR-006, FR-007, FR-008, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Authenticate the user normally, including MFA when required.
2	After successful authentication, issue a secure persistent authentication token.
3	Store no password on the client device.
4	On a later visit, validate the persistent token server-side.
5	Verify account status and security policy before establishing a session.
6	Require full/step-up authentication for sensitive actions when required.
7	Rotate or revoke persistent tokens according to policy.
8	Record security-relevant trusted-login events.

Alternative / Exception Flows

- Token is expired, revoked, or tampered with: reject and require full login.
- Account becomes locked/disabled: trusted login cannot bypass the state.
- Password change or Logout All Sessions: invalidate affected persistent tokens.
- Suspicious activity: revoke trust and require full authentication.

Postconditions

- Remember Me improves re-authentication convenience but never creates an unlimited session.

Security & Enterprise Controls

- Secure persistent cookies; token rotation/revocation; no password storage; MFA cannot be bypassed when policy requires it.

UC-007 – Multi-Factor / Step-Up Authentication

Use Case ID	UC-007
Primary Actor(s)	Authorized User
Supporting Systems / Actors	Authentication Service; MFA Provider; Audit Service
Trigger	Primary authentication succeeds or a sensitive action requires step-up authentication.
Preconditions	User/session is eligible for the protected operation.
Related Functional Requirements	FR-001, FR-008, FR-010, FR-031
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Determine whether MFA/step-up is required for the user, role, risk, or action.
2	Present the approved additional authentication challenge.
3	Receive the additional factor.
4	Validate the factor and its validity period.
5	On success, record the stronger authentication state.
6	Continue only with the operation permitted by policy.
7	Record the MFA result in the audit trail.

Alternative / Exception Flows

- Incorrect factor: reject the attempt.
- Expired factor: reject and allow a new challenge when permitted.
- Maximum attempts exceeded: apply security controls.
- MFA service unavailable: fail closed for operations that require MFA.
- Direct URL/API bypass attempt: reject server-side.

Postconditions

- The protected operation proceeds only after the required assurance level is satisfied.

Security & Enterprise Controls

- Step-up for sensitive operations; replay resistance; bounded attempts; server-side enforcement; audit logging.

UC-008 – Concurrent Session Management

Use Case ID	UC-008
Primary Actor(s)	Authenticated User
Supporting Systems / Actors	Session Management Service; Audit Service
Trigger	A new session is created or the user manages active sessions.
Preconditions	User identity is authenticated.
Related Functional Requirements	FR-004, FR-006, FR-009, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Identify existing active sessions for the account.
2	Apply the configured concurrent-session policy.
3	Create or deny the new session according to policy.
4	Allow the user to view permitted active-session information.
5	Allow termination of an unknown/old session.
6	Invalidate selected sessions server-side.
7	Support Logout All Sessions when permitted.
8	Record significant session creation and termination events.

Alternative / Exception Flows

- Concurrent-session limit exceeded: apply the configured deny/replace behavior.
- Revoked session token is reused: reject it.
- Suspicious session is detected: revoke or escalate according to security policy.

Postconditions

- Only sessions permitted by policy remain valid.

Security & Enterprise Controls

- Server-side session registry; token revocation; device/session metadata minimization; audit logging.

UC-009 – Role-Based Access Control (RBAC)

Use Case ID	UC-009
Primary Actor(s)	Authenticated User
Supporting Systems / Actors	Authorization Service; Audit Service
Trigger	User requests a protected function or resource.
Preconditions	User is authenticated; authorization policy is available.
Related Functional Requirements	FR-008, FR-010, FR-031
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Resolve the user role(s), permissions, and organizational/data scope.
2	Receive a request for a protected action/resource.
3	Evaluate authorization server-side.
4	Apply row/object/department/institution scope as applicable.
5	Invoke step-up authentication when the authorized action is sensitive.
6	Execute the request only when permitted.
7	Record sensitive/denied events as required.

Alternative / Exception Flows

- Permission missing: deny access.
- URL/API manipulation: reject independently of UI visibility.
- Cross-scope data request: deny unauthorized records.
- Mandatory authorization dependency unavailable: fail closed.

Postconditions

- No protected operation executes outside the user's effective permissions and scope.

Security & Enterprise Controls

- Default deny; least privilege; server/API enforcement; segregation of duties; auditable privilege use.

UC-010 – Student Course Registration

Use Case ID	UC-010
Primary Actor(s)	Student
Supporting Systems / Actors	Registration Engine; Academic Rules Service; SIS; Audit/Notification Services
Trigger	Student confirms registration for a selected section.
Preconditions	Student is authenticated; term/section are available for registration.
Related Functional Requirements	FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019, FR-027, FR-031, FR-033, FR-034
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Open Registration and select the academic term.
2	Search/select a course section.
3	Validate registration window and student academic status.
4	Validate prerequisites/co-requisites, holds, schedule conflicts, and credit load.
5	Validate seat availability.
6	Revalidate commit-critical rules at transaction commit.
7	Create the enrollment and update capacity atomically.
8	Prevent duplicate effects through idempotency.
9	Record the registration event.
10	Display confirmation and trigger the configured notification.

Alternative / Exception Flows

- Any mandatory academic validation fails: block and return a clear business reason.
- Registration window is closed: reject unless a valid approved exception applies.
- Last seat is concurrently claimed: only one commit succeeds.
- Duplicate/retried request: return the prior result without duplicate enrollment.

- Mandatory authoritative dependency unavailable: fail safe and do not guess eligibility.

Postconditions

- Exactly one valid enrollment is committed and capacity remains consistent, or no enrollment is created.

Security & Enterprise Controls

- Transactional commit; atomic capacity control; idempotency; server-side validation; complete audit correlation.

UC-011 – Course Search & Section Availability

Use Case ID	UC-011
Primary Actor(s)	Student
Supporting Systems / Actors	Course Catalog; Section Service
Trigger	Student searches for courses/sections.
Preconditions	A published term and course-section data are available.
Related Functional Requirements	FR-012, FR-018, FR-030, FR-034
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Open Course Search and select a term.
2	Search by permitted criteria such as course code, title, or department.
3	Apply optional filters.
4	Return only published/visible course sections.
5	Show instructor, schedule, location/delivery mode, capacity, and current availability where permitted.
6	Open section details and allow progression to registration when eligible.

Alternative / Exception Flows

- No matches: show an empty result with guidance.
- Section is unpublished: exclude it from registration results.
- Availability data is stale/unavailable: do not present a guessed seat count.
- Availability changes after search: registration commit performs a fresh capacity check.

Postconditions

- Search does not reserve a seat; displayed availability is informational until commit.

Security & Enterprise Controls

- Input validation; scoped visibility; freshness indicators where needed; no sensitive data leakage.

UC-012 – Registration Eligibility Validation

Use Case ID	UC-012
Primary Actor(s)	Student
Supporting Systems / Actors	Registration Rules Engine; SIS
Trigger	Student selects a section or requests validation.
Preconditions	Student and section identities are valid; policy data is available.
Related Functional Requirements	FR-013, FR-014, FR-015, FR-016, FR-017, FR-029, FR-033, FR-034
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Load the effective registration policy version for the term.
2	Validate registration window/deadlines.
3	Validate student academic status and registration holds.
4	Validate prerequisites and co-requisites.
5	Validate schedule conflicts.
6	Validate current/minimum/maximum credit-load rules.
7	Evaluate other applicable approved academic policies.
8	Return an eligibility decision and explicit reasons.
9	Carry approved exception references forward when valid.

Alternative / Exception Flows

- Any mandatory validation fails: block registration with the applicable reason.
- Required authoritative data is unavailable: fail closed.
- Approved exception exists: apply only within its scope and validity period.

Postconditions

- Eligibility is based on traceable effective rules and authoritative data.

Security & Enterprise Controls

- Deterministic rule evaluation; effective-dated policies; fail-safe dependencies; auditable overrides.

UC-013 – Seat Allocation & Concurrent Registration

Use Case ID	UC-013
Primary Actor(s)	Student
Supporting Systems / Actors	Registration Service; Database
Trigger	Student commits an enrollment request.
Preconditions	Eligibility checks have passed sufficiently to attempt commit.
Related Functional Requirements	FR-011, FR-018, FR-019, FR-031
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Receive the commit request with an idempotency/correlation key.
2	Recheck current seat capacity within the transaction boundary.
3	Check for existing enrollment or prior identical request.
4	Atomically allocate capacity and create enrollment.
5	Commit the transaction.
6	Return the committed result and record the event.

Alternative / Exception Flows

- Two students compete for the final seat: exactly one transaction succeeds.
- Duplicate request: return the original outcome without additional side effects.
- Commit fails: roll back enrollment/capacity changes.
- Section becomes full: return Section Full or route to waitlist if eligible.

Postconditions

- Capacity never falls below zero and duplicate enrollments are not created.

Security & Enterprise Controls

- Atomic transaction; concurrency control; unique constraints; idempotency; correlation IDs.

UC-014 – Add / Drop / Withdraw Course

Use Case ID	UC-014
Primary Actor(s)	Student
Supporting Systems / Actors	Registration Service; Academic/Financial Services; Notification Service
Trigger	Student selects Add, Drop, or Withdraw.
Preconditions	Student has an active academic record and the target action is applicable.
Related Functional Requirements	FR-020, FR-022, FR-023, FR-024, FR-027, FR-034
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Open the current schedule.
2	Select Add, Drop, or Withdraw and the target section.
3	Determine the current academic deadline and applicable policy.
4	Validate the action and show material consequences when required.
5	Obtain user confirmation.
6	Commit the enrollment-status change transactionally.
7	Update capacity and schedule state as applicable.
8	Trigger financial recalculation/posting when required.
9	Record the event and send confirmation/notification.

Alternative / Exception Flows

- Deadline passed: reject or route to an approved exception workflow when permitted.
- Action violates an academic rule: block with reason.
- Downstream noncritical notification fails: retain the committed business transaction and retry notification.
- Critical transaction component fails: roll back to a consistent state.

Postconditions

- Enrollment, capacity, schedule, and financial consequences are traceable and consistent.

Security & Enterprise Controls

- Transactional integrity; deadline enforcement; authorized exceptions; audit correlation.

UC-015 – Waitlist Management

Use Case ID	UC-015
Primary Actor(s)	Student
Supporting Systems / Actors	Registration System; Notification Service
Trigger	A section is full and waitlisting is enabled, or capacity becomes available.
Preconditions	Student is eligible to join the waitlist.
Related Functional Requirements	FR-018, FR-021, FR-027, FR-034
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Offer Join Waitlist for an eligible full section.
2	Validate the student against applicable waitlist rules.
3	Add the student once using the governed priority/order.
4	When capacity becomes available, select the next eligible candidate.
5	Revalidate eligibility before promotion/offer.
6	Promote or create an offer according to policy.
7	Update capacity and waitlist state atomically.
8	Notify the student and record the event.

Alternative / Exception Flows

- Student is already waitlisted: do not create a duplicate entry.
- Candidate is no longer eligible: skip/hold according to policy.
- Offer expires: advance to the next eligible candidate.
- Concurrent seat activity: concurrency control prevents oversubscription.

Postconditions

- Waitlist order and promotions are policy-driven, reproducible, and auditable.

Security & Enterprise Controls

- Atomic promotion; governed ordering; duplicate prevention; notification and audit trail.

UC-016 – Schedule & Registration Status Synchronization

Use Case ID	UC-016
Primary Actor(s)	Student / System
Supporting Systems / Actors	Registration Service; SIS; Integration Platform
Trigger	A registration status changes.
Preconditions	A committed source transaction exists.
Related Functional Requirements	FR-011, FR-020, FR-021, FR-022, FR-031, FR-035
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Capture the committed registration event.
2	Update the student schedule and registration status in the source-of-truth domain.
3	Publish the change to approved downstream integrations.
4	Propagate the correlation ID and version metadata.
5	Apply idempotency to repeated events.
6	Monitor delivery and reconcile target-system state.
7	Record synchronization outcomes.

Alternative / Exception Flows

- Downstream integration fails after a valid registration commit: preserve the valid source transaction and retry/reconcile.
- Duplicate event: apply once.
- Conflicting downstream state: resolve using authoritative ownership/version rules.

Postconditions

- Downstream systems converge on the authoritative registration state without duplicate side effects.

Security & Enterprise Controls

- Source-of-truth ownership; idempotency; retry/backoff; reconciliation; correlation logging.

UC-017 – Tuition & Fee Calculation

Use Case ID	UC-017
Primary Actor(s)	Student / System
Supporting Systems / Actors	Finance Rules Service
Trigger	A financially relevant enrollment/status event occurs.
Preconditions	Effective fee rules and student classification data are available.
Related Functional Requirements	FR-020, FR-023, FR-026, FR-029, FR-034
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Load effective tuition and fee rules for the term.
2	Determine applicable student classification and enrolled units/courses.
3	Calculate tuition and mandatory/conditional fees.
4	Apply authorized scholarships, waivers, or discounts.
5	Calculate the net amount due.
6	Persist the calculation basis, rule/version references, and result.
7	Present the financial summary to authorized users.

Alternative / Exception Flows

- Required fee configuration is missing: stop and do not guess an amount.
- Rule is not effective for the term: exclude it.
- Repeated financial event: avoid duplicate charging.
- A later rule change occurs: preserve historical calculation traceability.

Postconditions

- The calculated amount is reproducible from effective rules and source data.

Security & Enterprise Controls

- Effective-dated rates; deterministic calculation; idempotency; least-privilege financial data access.

UC-018 – Invoice & Student Account Posting

Use Case ID	UC-018
Primary Actor(s)	Student / Finance System
Supporting Systems / Actors	Student Account Service; Integration Platform
Trigger	An approved financial calculation is ready for posting.
Preconditions	A traceable fee calculation or adjustment exists.
Related Functional Requirements	FR-023, FR-024, FR-026, FR-031, FR-035
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Receive the approved charge/credit result.
2	Create a unique student-account transaction.
3	Generate/update the invoice or statement as applicable.
4	Post the amount to the student account.
5	Synchronize the posting with the authoritative financial system.
6	Expose the permitted balance/statement to the student.
7	Record transaction and integration identifiers.

Alternative / Exception Flows

- Duplicate posting request: return the original posting without another ledger effect.
- Financial system unavailable: retain controlled pending/reconciliation state rather than fabricating success.
- Reconciliation mismatch: flag for governed resolution.

Postconditions

- Student-account and invoice states are balanced, unique, and traceable.

Security & Enterprise Controls

- Ledger integrity; idempotency; reconciliation; protected financial data; audit logging.

UC-019 – Online Payment Processing

Use Case ID	UC-019
Primary Actor(s)	Student
Supporting Systems / Actors	Payment Gateway; Student Account Service; Notification Service
Trigger	Student selects Pay for an eligible balance/invoice.
Preconditions	Amount due exists; payment service is configured and available.
Related Functional Requirements	FR-024, FR-025, FR-027, FR-031, FR-035
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Display the eligible amount/invoice.
2	Create a unique payment request and correlation ID.
3	Transfer only necessary payment context to the approved provider.
4	Allow the student to complete payment with the provider.
5	Receive and authenticate the provider response/callback.
6	Update payment status and account balance only from trusted confirmed results.
7	Issue a receipt when successful.
8	Record the transaction and send confirmation.

Alternative / Exception Flows

- Payment declined: retain unpaid state.
- Timeout/unknown result: use Pending/Reconciliation, not Success.
- Duplicate callback: apply idempotently.
- Malformed/unauthorized callback: reject and alert/log.

Postconditions

- Successful payment is posted once; ambiguous payment remains reconcilable.

Security & Enterprise Controls

- Tokenized/hosted payment handling; no unnecessary card data storage; callback authentication; idempotency; audit.

UC-020 – Scholarships, Waivers & Discounts

Use Case ID	UC-020
Primary Actor(s)	Authorized Finance Staff / System
Supporting Systems / Actors	Financial Aid Service; Finance Rules Service
Trigger	Eligible aid/discount data or an authorized adjustment is available.
Preconditions	Actor/data source is authorized; effective policy exists.
Related Functional Requirements	FR-023, FR-024, FR-026, FR-028, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Retrieve approved aid/discount/waiver data.
2	Validate student eligibility, scope, amount/percentage, and effective dates.
3	Apply the authorized benefit to the applicable charges.
4	Recalculate the net balance.
5	Persist source, rule, reason, and approval references.
6	Record the financial adjustment.

Alternative / Exception Flows

- Eligibility fails or benefit expired: do not apply.
- Manual override lacks permission/approval: reject.
- Conflicting benefits exceed policy limits: apply governed precedence/cap rules.

Postconditions

- Only authorized, traceable benefits affect the student balance.

Security & Enterprise Controls

- Least privilege; approval for material overrides; step-up MFA where configured; before/after audit values.

UC-021 – Notifications & Communications

Use Case ID	UC-021
Primary Actor(s)	System
Supporting Systems / Actors	Notification Service; Email/SMS/Portal Channels
Trigger	A configured business/security event requires communication.
Preconditions	A valid event, recipient, template, and permitted channel are available.
Related Functional Requirements	FR-027, FR-031, FR-035
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Receive the business/security event.
2	Resolve the approved template, recipient, locale/preferences, and channel.
3	Minimize sensitive data included in the message.
4	Render the communication.
5	Send through the approved channel.
6	Capture provider delivery status.
7	Retry transient failures according to policy.
8	Record communication metadata and correlation ID.

Alternative / Exception Flows

- Invalid recipient/channel: suppress and flag for resolution.
- Transient delivery failure: retry with bounded backoff.
- Permanent failure: record final failure and route as configured.
- Notification failure after a committed business transaction: do not reverse that transaction unless explicitly required.

Postconditions

- Communication delivery status is traceable without exposing unnecessary sensitive content.

Security & Enterprise Controls

- Template governance; data minimization; opt/preference rules where applicable; secure provider integration; audit.

UC-022 – Exception & Approval Workflow

Use Case ID	UC-022
Primary Actor(s)	Student / Authorized Requester / Approver
Supporting Systems / Actors	Workflow Service; Audit Service
Trigger	A permitted exception/override is requested.
Preconditions	The underlying policy supports governed exceptions.
Related Functional Requirements	FR-008, FR-010, FR-028, FR-029, FR-031
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Create an exception request with reason and required evidence.
2	Validate request completeness and requester authority.
3	Route to the configured approver(s).
4	Enforce segregation of duties and self-approval restrictions.
5	Require step-up MFA for sensitive decisions when configured.
6	Approver reviews and approves/rejects with rationale.
7	Record decision, scope, validity period, and approval references.
8	Apply an approved exception only within its defined scope.
9	Notify relevant parties.

Alternative / Exception Flows

- Missing evidence: return for correction.
- Unauthorized/self-approval attempt: reject.
- Request expires or underlying condition changes: revalidate before use.
- Rejected request: do not alter the governing rule or transaction.

Postconditions

- Approved exceptions are bounded and auditable; rejected/expired exceptions have no effect.

Security & Enterprise Controls

- Segregation of duties; step-up MFA; scoped/expiring overrides; immutable decision history.

UC-023 – Academic Policy & Rule Administration

Use Case ID	UC-023
Primary Actor(s)	Academic Policy Administrator / Registrar
Supporting Systems / Actors	Policy Repository; Approval Workflow; Audit Service
Trigger	Authorized administrator creates or changes a rule.
Preconditions	Actor has explicit policy-administration permission.
Related Functional Requirements	FR-010, FR-013, FR-014, FR-015, FR-016, FR-017, FR-028, FR-029, FR-031
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Create a draft rule/version with rationale and effective period.
2	Validate structure, references, effective dates, and conflicts.
3	Test the draft against controlled scenarios.
4	Route material changes for configured approval.
5	Activate only an approved version at its effective date.
6	Retain prior versions and their historical applicability.
7	Retire/rollback through a governed subsequent version.
8	Record complete before/after and approval history.

Alternative / Exception Flows

- Conflicting/overlapping active rules: block activation.
- Emergency change: use the defined emergency-governance path with retrospective review.
- Unauthorized administrator: deny server-side.

Postconditions

- Production transactions use only approved effective rule versions.

Security & Enterprise Controls

- Default deny; dual control where required; change governance; step-up MFA; complete version/audit history.

UC-024 – Course Section & Offering Administration

Use Case ID	UC-024
Primary Actor(s)	Registrar / Department Staff / Academic Administrator
Supporting Systems / Actors	Course Catalog; Scheduling Services; Audit Service
Trigger	Authorized staff creates or modifies a term section.
Preconditions	Course/term and referenced entities exist in authoritative sources.
Related Functional Requirements	FR-010, FR-012, FR-018, FR-021, FR-027, FR-029, FR-030, FR-031, FR-034, FR-035
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Create or clone a term section from an approved course.
2	Assign section number, delivery mode, campus/location, meetings, instructor(s), capacity, and registration attributes.
3	Validate meeting consistency, capacity, dates, and references.
4	Publish only after required data/approvals are satisfied.
5	Expose the section to search/registration according to calendar rules.
6	Version/audit material post-publication changes.
7	Trigger impact analysis and notifications for affected enrollments when necessary.

Alternative / Exception Flows

- Invalid room/instructor/reference: block publication or flag existing offering for resolution.
- Material time/capacity change with active enrollment: perform impact analysis; do not silently move/remove students.
- Unauthorized department scope: deny.

Postconditions

- Published section configuration is authoritative for registration controls and historically traceable.

Security & Enterprise Controls

- Department/institution scope; controlled publication; impact analysis; before/after audit.

UC-025 – Audit Trail & Compliance Monitoring

Use Case ID	UC-025
Primary Actor(s)	Auditor / IT Security Manager / Authorized Operations
Supporting Systems / Actors	Audit Platform; Monitoring Service
Trigger	An auditable event occurs or an authorized reviewer queries audit records.
Preconditions	Audit service is available or supports durable protected buffering.
Related Functional Requirements	FR-001–FR-035; FR-031 primary
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Capture event category, actor/subject, action, resource, result, timestamp, correlation ID, and relevant rule/approval/before-after data.
2	Protect audit records from ordinary application modification/deletion.
3	Allow authorized search/filter by supported dimensions.
4	Allow restricted export under role/data-classification controls.
5	Detect configured anomalous patterns and raise monitoring events.
6	Retain/archive records according to approved policy.
7	Audit access to audit data itself where required.

Alternative / Exception Flows

- Audit sink temporarily unavailable: use durable protected buffering; critical actions may fail closed where non-repudiation is mandatory.
- Unauthorized audit query/export: deny and record the attempt.
- Export scope exceeds authorization: deny or redact according to policy.

Postconditions

- Auditable activity remains reconstructable and tamper-resistant for the required retention period.

Security & Enterprise Controls

- Append-only application perspective; access segregation; encryption; time synchronization; restricted export; correlation.

UC-026 – Reporting & Management Dashboards

Use Case ID	UC-026
Primary Actor(s)	Registrar / Department Head / Finance Manager / Management / Analyst
Supporting Systems / Actors	Reporting Platform; Governed Data Sources
Trigger	Authorized user opens a report/dashboard.
Preconditions	Reporting data is synchronized and the user has report/data scope.
Related Functional Requirements	FR-010, FR-018, FR-021, FR-023, FR-026, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035
Priority	High

Main Success Flow

Step	System / Actor Behavior
1	Select an approved report/dashboard and permitted organizational/term scope.
2	Apply supported filters.
3	Compute/display metrics from governed definitions.
4	Display data-freshness information.
5	Enforce row/department/institution data scope.
6	Allow authorized export with classification-aware controls.
7	Record governed report/export metadata as required.

Alternative / Exception Flows

- Reporting data is stale: show a freshness warning and restrict decisions that require real-time state.
- Unauthorized cross-department request: exclude/deny data.
- Large export: process securely with controlled retrieval/expiry.

Postconditions

- Reports preserve source authorization and clearly communicate data freshness.

Security & Enterprise Controls

- Row-level security; data minimization; export controls; formula/CSV-injection protection where applicable; audit logging.

UC-027 – Student Profile & Academic Status Management

Use Case ID	UC-027
Primary Actor(s)	Student / Registrar Staff / Academic Advisor
Supporting Systems / Actors	Authoritative SIS; Audit Service
Trigger	Student profile/status is requested or synchronized.
Preconditions	Student identity is matched to an authoritative institutional record.
Related Functional Requirements	FR-010, FR-011, FR-016, FR-020, FR-023, FR-026, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-035
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Retrieve/synchronize registration-relevant identity, program, department, level, academic standing, classification, and status fields.
2	Expose only fields permitted for the actor.
3	Allow self-service maintenance only for designated non-authoritative fields.
4	Route authoritative academic-data corrections to the owning system/process.
5	Expose current status to registration rules with freshness/version metadata.
6	Propagate material status changes for revalidation of affected pending/conditional transactions.
7	Record sensitive profile/status events.

Alternative / Exception Flows

- Identity mismatch/duplicate source record: quarantine for data-quality resolution.
- Authoritative SIS unavailable: fail closed for rules requiring unavailable status.
- Student attempts to alter protected academic status: deny with no source change.

Postconditions

- Registration decisions consume authoritative, appropriately scoped student status.

Security & Enterprise Controls

- Field-level access; privacy minimization; authoritative ownership; server-side authorization; data-quality audit.

UC-028 – Academic Term & Calendar Management

Use Case ID	UC-028
Primary Actor(s)	Registrar / Academic Calendar Administrator
Supporting Systems / Actors	Calendar Repository; Approval/Audit Services
Trigger	Authorized user creates or amends an academic term/calendar.
Preconditions	Actor has calendar-administration permission and approved institutional inputs.
Related Functional Requirements	FR-010, FR-011, FR-012, FR-017, FR-020, FR-021, FR-022, FR-023, FR-027, FR-029, FR-030, FR-031, FR-032, FR-034, FR-035
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Create term identity, start/end dates, status, and lifecycle milestones.
2	Configure population-specific registration windows and action deadlines with timezone.
3	Configure add/drop/withdraw deadlines, holidays, and related periods.
4	Validate date sequence, overlap, term references, and policy consistency.
5	Route/publish the approved calendar version for operational use.
6	Apply trusted server time to deadline decisions.
7	Version subsequent amendments and trigger impact assessment/notifications.
8	Record change rationale and before/after values.

Alternative / Exception Flows

- Conflicting term/window dates: block publication.
- Emergency closure/deadline extension: use a governed amendment with explicit scope/reason.
- Term closure has unresolved critical processes: block unless an authorized closure procedure handles them.

Postconditions

- Operational processes reference an approved, effective, timezone-aware calendar version.

Security & Enterprise Controls

- Institution-scoped change control; trusted time; versioning; optional dual approval; complete audit.

UC-029 – Enterprise Integration & Data Synchronization

Use Case ID	UC-029
Primary Actor(s)	System / Authorized Integration Services
Supporting Systems / Actors	SIS; Identity Provider; Finance/Payment; Notification; Reporting/Integration Platforms
Trigger	An approved inbound/outbound integration transaction or synchronization event occurs.
Preconditions	Integration contract, service identity, endpoint, ownership, and data classification are defined.
Related Functional Requirements	FR-019, FR-022, FR-024, FR-025, FR-027, FR-031, FR-033, FR-035
Priority	Critical

Main Success Flow

Step	System / Actor Behavior
1	Authenticate service-to-service using managed credentials and least-privilege scopes.
2	Validate request/response schema, identifiers, version, and correlation metadata.
3	Determine source-of-truth ownership and allowed synchronization direction.
4	Apply idempotency and ordering/version rules before mutation.
5	Process the transaction and propagate the correlation ID end-to-end.
6	Retry transient failures with bounded backoff.
7	Route permanent/poison failures to controlled exception handling.
8	Monitor latency, failures, reconciliation state, and contract/version health.
9	Record integration outcomes without logging secrets.

Alternative / Exception Flows

- Malformed or unauthorized payload: reject and record the security/monitoring event.
- Duplicate event: apply once with no duplicate side effects.
- Conflicting update: authoritative ownership/version rules resolve or quarantine it.

- Dependency outage: use the specifically approved fail-closed, queued, or degraded behavior; never fabricate authoritative data.

Postconditions

- Integrated systems converge according to explicit ownership, version, and reconciliation rules.

Security & Enterprise Controls

- TLS; managed secret rotation; service RBAC; schema validation; replay protection; rate controls; observability; reconciliation; audit correlation.

Traceability Coverage Summary

The following matrix provides the primary UC-to-FR references used in this master. Cross-cutting controls such as authorization and audit may apply more broadly than the explicit minimum references shown in individual flows.

Use Case	Related Functional Requirements
UC-001 – Student Login & Authentication	FR-001, FR-005, FR-006, FR-008, FR-009, FR-010, FR-031
UC-002 – Forgot Password / Password Recovery	FR-002, FR-005, FR-008, FR-027, FR-031
UC-003 – Change Password	FR-003, FR-008, FR-010, FR-027, FR-031
UC-004 – Secure User Logout	FR-004, FR-006, FR-007, FR-009, FR-031
UC-005 – Session Timeout & Automatic Logout	FR-004, FR-006, FR-031
UC-006 – Remember Me / Trusted Login	FR-001, FR-004, FR-005, FR-006, FR-007, FR-008, FR-031
UC-007 – Multi-Factor / Step-Up Authentication	FR-001, FR-008, FR-010, FR-031
UC-008 – Concurrent Session Management	FR-004, FR-006, FR-009, FR-031
UC-009 – Role-Based Access Control (RBAC)	FR-008, FR-010, FR-031
UC-010 – Student Course Registration	FR-010, FR-011, FR-012, FR-013, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019, FR-027, FR-031, FR-033, FR-034
UC-011 – Course Search & Section Availability	FR-012, FR-018, FR-030, FR-034
UC-012 – Registration Eligibility Validation	FR-013, FR-014, FR-015, FR-016, FR-017, FR-029, FR-033, FR-034
UC-013 – Seat Allocation & Concurrent Registration	FR-011, FR-018, FR-019, FR-031
UC-014 – Add / Drop / Withdraw Course	FR-020, FR-022, FR-023, FR-024, FR-027, FR-034
UC-015 – Waitlist Management	FR-018, FR-021, FR-027, FR-034
UC-016 – Schedule & Registration Status Synchronization	FR-011, FR-020, FR-021, FR-022, FR-031, FR-035
UC-017 – Tuition & Fee Calculation	FR-020, FR-023, FR-026, FR-029, FR-034

Use Case	Related Functional Requirements
UC-018 – Invoice & Student Account Posting	FR-023, FR-024, FR-026, FR-031, FR-035
UC-019 – Online Payment Processing	FR-024, FR-025, FR-027, FR-031, FR-035
UC-020 – Scholarships, Waivers & Discounts	FR-023, FR-024, FR-026, FR-028, FR-031
UC-021 – Notifications & Communications	FR-027, FR-031, FR-035
UC-022 – Exception & Approval Workflow	FR-008, FR-010, FR-028, FR-029, FR-031
UC-023 – Academic Policy & Rule Administration	FR-010, FR-013, FR-014, FR-015, FR-016, FR-017, FR-028, FR-029, FR-031
UC-024 – Course Section & Offering Administration	FR-010, FR-012, FR-018, FR-021, FR-027, FR-029, FR-030, FR-031, FR-034, FR-035
UC-025 – Audit Trail & Compliance Monitoring	FR-001–FR-035; FR-031 primary
UC-026 – Reporting & Management Dashboards	FR-010, FR-018, FR-021, FR-023, FR-026, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035
UC-027 – Student Profile & Academic Status Management	FR-010, FR-011, FR-016, FR-020, FR-023, FR-026, FR-028, FR-029, FR-030, FR-031, FR-032, FR-033, FR-035
UC-028 – Academic Term & Calendar Management	FR-010, FR-011, FR-012, FR-017, FR-020, FR-021, FR-022, FR-023, FR-027, FR-029, FR-030, FR-031, FR-032, FR-034, FR-035
UC-029 – Enterprise Integration & Data Synchronization	FR-019, FR-022, FR-024, FR-025, FR-027, FR-031, FR-033, FR-035

Functional Coverage Confirmation

Collectively, UC-001 through UC-029 cover the Enterprise functional baseline FR-001 through FR-035. Detailed RTM work should next connect Business Requirements, Business Rules, FRs, NFRs, Use Cases, Acceptance Criteria, and Test Cases without renumbering this baseline.